

Electrical Parameter Monitoring System Using ESP32

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Abstract—Monitoring electrical parameters such as voltage, current, power, and power factor is essential for efficient energy management in industrial and domestic environments. Traditional monitoring systems often require complex hardware and expensive instruments, making them less suitable for small-scale applications. This project presents a simple and cost-effective electrical parameter monitoring system using the ESP32 microcontroller and the PZEM004T energy monitoring module. The PZEM004T measures voltage, current, power, and power factor of a connected load, with a current transformer used for current sensing. The ESP32 communicates with the PZEM004T through serial communication to obtain the measured data. These parameters are then processed and displayed in real time on a 16×2 LCD using the I2C communication protocol. The system displays readings locally on the LCD without relying on cloud platforms or mobile dashboards. The proposed system is compact, reliable, and suitable for educational and small-scale industrial monitoring applications.

Keywords—ESP32, Energy Monitoring, PZEM004T, Electrical Parameters, Power Monitoring, I2C LCD, Embedded System

I. INTRODUCTION

Efficient monitoring of electrical parameters such as voltage, current, power, and power factor is essential for effective energy management in residential, commercial, and industrial environments. With the increasing demand for electricity and the need to reduce energy wastage, real-time monitoring systems have become an important part of modern electrical infrastructure. [1]. Traditional measurement instruments are often bulky, expensive, and require manual observation, which makes continuous monitoring difficult.

Recent developments in embedded systems and Internet of Things (IoT) technologies have enabled the design of compact and low-cost energy monitoring solutions. Microcontrollers such as the ESP32 provide powerful processing capability, built-in communication interfaces, and low power consumption, making them suitable for smart monitoring applications. In addition, dedicated energy monitoring modules such as the PZEM004T allow accurate measurement of electrical parameters including voltage, current, power, and power factor. [2].

This project presents the design and implementation of a real-time electrical parameter monitoring system using the ESP32 microcontroller and the PZEM004T energy monitoring module. The system collects electrical measurements from the PZEM004T and displays the values on a 16×2 LCD using the I2C communication interface. [3]. By providing real-time information about power consumption, the proposed system helps users understand energy usage patterns and supports better energy management.

The developed system is simple, reliable, and cost-effective, making it suitable for educational purposes as well as small-scale monitoring applications in homes and laboratories. [4].

II. LITERATURE SURVEY / STATE OF THE ART TECHNOLOGY

Electrical energy monitoring systems are widely used in industrial and domestic environments to measure important electrical parameters such as

voltage, current, power, and power factor. Monitoring these parameters helps in improving energy efficiency and managing power consumption effectively. Traditionally, electrical measurements were performed using analog instruments such as voltmeters, ammeters, and wattmeters, which required manual observation and could not provide continuous monitoring [5].

With advancements in embedded systems, microcontroller-based energy monitoring systems have become more common. Microcontrollers such as Arduino, PIC, and ESP32 are widely used for monitoring applications because they can easily interface with sensors and display devices to provide real-time measurement and analysis [6].

Modern monitoring systems often use dedicated energy measurement modules such as the PZEM004T, which can measure voltage, current, power, and power factor with high accuracy. These modules simplify system design by integrating multiple measurement functions into a single device [7].

Microcontroller-based energy monitoring systems using such modules provide a compact, low-cost, and efficient solution for real-time electrical parameter monitoring in laboratories, educational projects, and small-scale industrial applications [8].

III. LIMITATIONS

1. High Cost Many commercial smart meters and industrial monitoring systems are expensive due to advanced sensing devices and additional hardware components. This makes them less suitable for small-scale applications and educational projects.

2. Complex System Architecture Modern monitoring systems often involve multiple components such as sensors, microcontrollers, and communication modules. This increases the overall system complexity and makes the design and implementation more difficult.

3. High Power Consumption Advanced monitoring systems that use multiple electronic modules may consume more power, which can reduce overall efficiency in long-term operation.

4. Difficult Installation and Maintenance Complex monitoring systems usually require skilled technicians for installation and maintenance, making them less convenient for general users.

5. Dependence on External Infrastructure

Some systems rely on external communication networks or centralized processing units. If these systems fail, monitoring may stop or become unreliable.

IV. PROPOSED SOLUTION / METHODOLOGY

A. PZEM004T Energy Monitoring Module

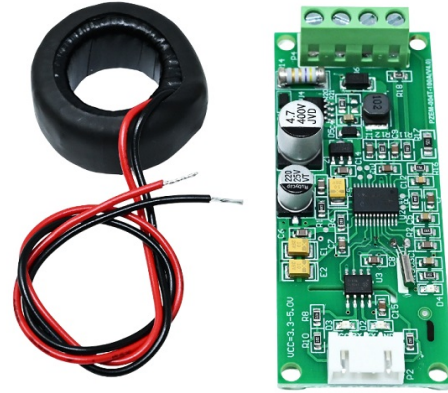


Fig. 1: PZEM Module with Current Transformer

The PZEM004T energy monitoring module is used to measure key electrical parameters such as voltage, current, power, and power factor of the connected electrical load. It directly measures the supply voltage through its input terminals and senses the current using an external current transformer (CT). The module internally processes these measurements and calculates real power and power factor using embedded measurement circuitry. The PZEM004T operates within a typical voltage range of 80–260 V AC and supports current measurement up to 100 A depending on the CT used. The measured parameters are transmitted to the ESP32 microcontroller through UART serial communication at a baud rate of 9600 bps, allowing the microcontroller to access accurate real-time electrical data.

B. Current Transformer (CT)

The current transformer (CT) is used for sensing the current flowing through the electrical load without directly exposing the measurement circuit to high current. The live wire supplying power to the load is passed through the CT sensor, which operates on the principle of electromagnetic induction. When current flows through the primary conductor, a proportional current is induced in the secondary winding of the transformer. This signal is then processed by the PZEM004T module to determine the actual current value. The use of a current transformer provides electrical isolation between the high-voltage power circuit and the measurement electronics, thereby improving safety and reliability in the monitoring system.

C. ESP32 Microcontroller

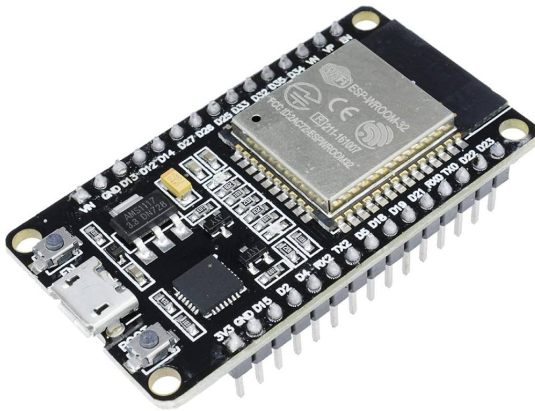


Fig. 2: ESP32 Microcontroller

The ESP32 microcontroller acts as the main processing unit of the monitoring system. It is a high-performance microcontroller developed by Espressif Systems and features a dual-core processor operating up to 240 MHz, integrated Wi-Fi and Bluetooth capabilities, and multiple communication interfaces such as UART, SPI, and I2C. In this system, the ESP32 communicates with the PZEM004T module using UART serial communication to receive the measured electrical parameters. After acquiring the data, the ESP32 processes the values and prepares them for display on the LCD screen. Its high processing capability and versatile communication interfaces make it suitable for real-time embedded monitoring applications.

D. 16x2 LCD Display with I2C Interface



Fig. 3: 16x2 LCD Display with I2C Interface

The system uses a 16x2 Liquid Crystal Display (LCD) with an I2C interface to display the measured electrical parameters. The LCD can display 16 characters in two rows, which allows multiple parameters to be shown simultaneously. The I2C interface simplifies hardware connections by requiring only two communication lines: Serial Data (SDA) and Serial Clock (SCL). The ESP32 sends the processed electrical parameter values to the LCD through the I2C communication protocol. This enables real-time visualization of voltage, current, power, and power factor, providing a simple and user-friendly interface for monitoring the electrical system.

E. Electrical Load (Bulb)

In the proposed system, a bulb is used as the electrical load to demonstrate the functionality of the monitoring system. When the bulb is powered, electrical current flows through the circuit and passes through the current transformer connected to the PZEM004T module. This allows the module to measure the voltage across the load and the current flowing through it. Based on these measurements, the module calculates the power consumed by the load and the corresponding power factor. Since a



Fig. 4: Electrical Load (BULB)

bulb typically behaves as a resistive load, the measured power factor is generally close to unity (PF 1), making it suitable for testing and demonstration purposes.

V. OVERALL SYSTEM OPERATION

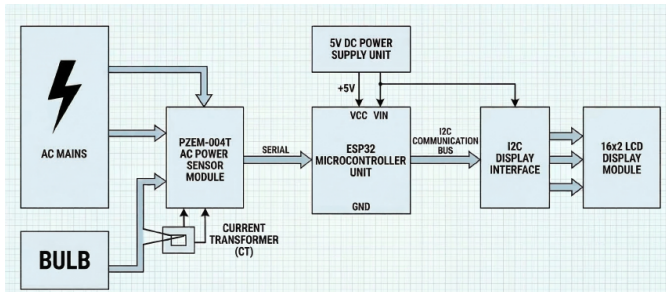
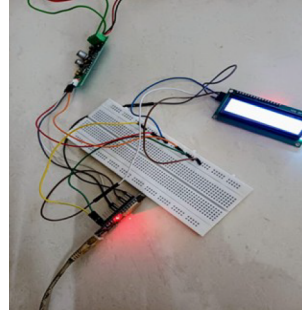


Fig. 5: Block Diagram of the system

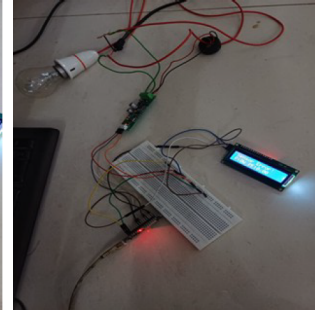
When the system is powered on, the ESP32 microcontroller initializes the LCD display and establishes serial communication with the PZEM004T energy monitoring module. The PZEM module continuously measures the voltage and current of the connected load using its internal measurement circuitry and the external current transformer. These measured values are transmitted to the ESP32 through UART communication. The ESP32 processes the received data and displays the voltage, current, power, and power factor on the LCD screen.

The readings are updated periodically, enabling real-time monitoring of electrical parameters and providing an efficient and compact solution for energy monitoring applications.

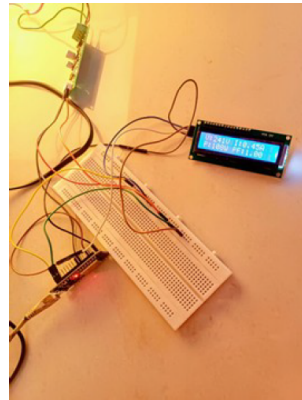
VI. RESULTS



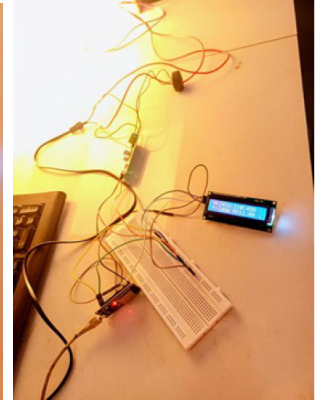
(a) Overall Circuit



(b) System Turned On



(a) Bulb and Its Readings



(b) Bulb and Its Readings

Fig. 7: Experimental Output of the Electrical Parameter Monitoring System

The system successfully measured electrical parameters of a connected bulb load. The observed values are:

- Voltage: 241.0 V
- Current: 0.45 A
- Power: 108 W
- Power Factor: 1.00 (for Filament Bulb)

VII. CONCLUSION

In this project, a real-time electrical parameter monitoring system was successfully developed using the ESP32 microcontroller and the PZEM004T energy monitoring module. The system is capable of measuring important electrical parameters such

as voltage, current, power, and power factor of a connected electrical load.

The ESP32 communicates with the PZEM004T module to obtain the measured values and displays them on a 16×2 I2C LCD display in real time. The use of the PZEM004T module simplifies the measurement process and provides accurate readings with minimal hardware complexity.

The developed system is simple, low-cost, and easy to implement, making it suitable for laboratory experiments, educational projects, and small-scale electrical monitoring applications. The system provides an efficient way to observe electrical parameters directly without requiring complex monitoring equipment.

Overall, the project demonstrates an effective solution for monitoring electrical parameters and can be further enhanced in the future by adding features such as data logging, remote monitoring, or integration with other monitoring systems.

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VIII. APPENDIX

```

1 // Libraries
2 #include <WiFi.h>
3 #include <PZEM004Tv30.h>
4 #include <LiquidCrystal_I2C.h>
5
6 // WiFi Credentials
7 char ssid[] = "Aa";
8 char pass[] = "12345678";
9
10 // Hardware Pins
11 #define PZEM_RX_PIN 16
12 #define PZEM_TX_PIN 17
13 #define PZEM_SERIAL Serial2
14
15 // Objects
16 PZEM004Tv30 pzem(PZEM_SERIAL, PZEM_RX_PIN
17     , PZEM_TX_PIN);
18
19 // LCD
20 LiquidCrystal_I2C lcd(0x27, 16, 2);
21
22 unsigned long lastMillis = 0;
23
24 void setup() {
25     Serial.begin(115200);
26     Serial.begin(115200);
27
28     // LCD SETUP
29     lcd.init();
30     lcd.backlight();
31
32     lcd.setCursor(0,0);
33     lcd.print("System Booting...");
34     delay(4000);
35
36     // WiFi Connection
37     lcd.clear();
38     lcd.print("WiFi Connecting...");
39
40     WiFi.begin(ssid, pass);
41
42     while (WiFi.status() != WL_CONNECTED) {
43         delay(4000);
44         Serial.print(".");
45     }
46     Serial.println("WiFi Connected");
47
48     lcd.clear();
49     lcd.setCursor(0,0);
50     lcd.print("WiFi Connected");
51
52     Serial.print("IP Address: ");
53     Serial.println(WiFi.localIP());
54
55     delay(6000);
56 }
57
58 void loop() {
59     if (millis() - lastMillis > 4000) {
60         lastMillis = millis();
61
62         // Read Data
63         float voltage = pzem.voltage();
64         float current = pzem.current();
65         float power = pzem.power();
66         float pf = pzem.pf();
67
68         // Serial Monitor Output
69         Serial.print("Voltage: ");
70         Serial.println(voltage);
71
72         Serial.print("Current: ");
73         Serial.println(current);
74
75         Serial.print("Power: ");
76         Serial.println(power);
77
78         Serial.print("Power Factor: ");
79         Serial.println(pf);
80
81         // LCD Display
82         lcd.clear();
83
84         lcd.setCursor(0,0);
85         lcd.print("V:");
86         lcd.print(voltage,0);
87         lcd.print("V ");
88
89         lcd.print("I:");
90         lcd.print(current,2);
91
92         lcd.print("A");
93         lcd.setCursor(0,1);
94
95         lcd.print("P:");
96         lcd.print(power,0);
97
98         lcd.print("W ");
99
100         lcd.print("PF:");
101         lcd.print(pf,2);
102
103     }
104 }
105

```